

SARAH SUSNEA

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EDUCATION

SMITH COLLEGE

A.B. in Statistical and Data Sciences, Minor in Sociology

Northampton, MA

September 2022 - May 2025

Cumulative GPA: 3.86/4.0

Relevant Coursework: Capstone in Statistical and Data Sciences, Advanced Programming for Data Science in R, Modeling for Machine Learning, Multiple Regression, Research Design and Analysis, Calculus III, Calculus II, Probability, Mathematical Statistics, Linear Algebra, Discrete Mathematics, SQL for Data Science, Qualitative Research Methods

WORK EXPERIENCE

MEMORIAL SLOAN KETTERING PCCTC

New York, NY (Remote)

Data Science Intern

June 2024 - August 2024

- Wrangled clinical trials data to create summary tables for safety and efficacy reports which adhered to regulatory standards
- Contributed a function for data validation and quality control to an internal package
- Built a dynamic HTML dashboard deployed to Posit Connect
- Programmatically interfaced with REST APIs to visualize real time data
- Submitted new code for review via pull requests on GitHub
- Contributed documentation on internal processes to a guide built in Quarto

WURTELE CENTER FOR LEADERSHIP

Northampton, MA

Pedagogical Partner

January 2025 - May 2025

- Selected for an exclusive fellowship based on demonstrated proficiency, experience, and passion for statistics, data science, and teaching
- Restructured data science ethics curriculum to strengthen connections between concepts and terminology in philosophy to data science projects within and beyond the classroom
- Developed worksheets and presentations in Quarto to facilitate student discussions on data science ethics and statistical definitions of fairness

SPINELLI CENTER FOR QUANTITATIVE LEARNING

Northampton, MA

Statistics and Data Sciences Tutor

September 2023 - May 2025

- Provided in class support for students as a teaching assistant in Professor Will Hopper's Introduction to Statistics class (Fall 2023, Fall 2024) and Multiple Regression class (Spring 2024)
- Provided drop in tutoring hours for Smith students once a week in Statistics and Data Science classes; responded to student questions in individual and group settings to guide them towards a deeper understanding of course material

RESEARCH EXPERIENCE

PEDAGOGICAL PARTNERSHIP

Northampton, MA

Undergraduate Research Assistant

January 2025 - May 2025

- Evaluated proposed methods for sports analytics through applying statistical definitions of fairness
- Developed an R package to facilitate the exploration of statistical definitions of fairness in machine learning
- Co-author on Baumer, B. S., & Susnea, S. B., 2025

SDS 410 - CAPSTONE IN STATISTICAL AND DATA SCIENCES

Northampton, MA

Undergraduate Research Assistant

September 2024 - December 2024

- Completed in partnership with Dr. Douglas Jones from UMASS Memorial Medical Center
- Applied machine learning methodology and statistical knowledge to define a novel metric to evaluate and compare vascular surgical quality across centers and surgeons
- Collaborated via Git repository by forking and cloning, issuing pull requests, and creating branches
- Listed as contributor on Jones et al., 2025

DISABILITY INCLUSION ANALYTICS LAB

Northampton, MA

Undergraduate Research Assistant

January 2024 - December 2024

- Conducted statistical analysis via regression models to analyze NHIS survey data
- Completed literature review focusing on accessibility and participatory mapping
- Co-author on Cao, S., Rosenfeld, H., & Susnea, S., 2025

PUBLICATIONS

- Jones, D. W., Simons, J. P., Markert, E., MacGibbon, C., Huang, J., Lin, N., **Susnea, S.**, Baumer, B. S., Scali, S., Stone, D. H., Schermerhorn, M., Schanzer, A. (2025). Earned Outcome Metrics Outperform Recommended Open Abdominal Aortic Aneurysm Repair Volume Guidelines in Discriminating Perioperative Mortality Among Centers. *Journal of Vascular Surgery*
- Cao, S., Rosenfeld, H., & **Susnea, S.** (2025). Educational opportunities of participatory GIS for accessibility on a college campus. *Journal of Geography in Higher Education*, 1–15.
- Baumer, B. S., & **Susnea, S. B.** (2025). What's the fairest of them all? (in progress)

AWARDS

- PositConf:: 2025 Opportunity Scholar
- Mu Sigma Rho 2025-2026
- 2025 Five College Statistics Prize - Smith College

SKILLS

Technical Skills: R, Git, SQL, Python, HTML, CSS, Microsoft Excel

Analysis Skills: Data processing and visualization, visual analytics, data dashboards, machine learning, linear and logistic regression, linear algebra, probability, mathematical statistics, computational statistics, statistical inference, web-scraping, regular expressions